

# Paper III v1.4 Internal Audit Final Report

Internal author-side audit — not independent

22 August 2026 — Overall verdict: PASS

**Protocol:** INTERNAL\_AUDIT\_STANDARD\_v1.3 plus Paper III v1.4 regressions.

**Lean execution:** recorded build evidence reviewed; full build not rerun.

## 1 Frozen target

Formal archive: PAPER\_III\_lean\_v1.4\_freeze.zip.

SHA-256:

79ee24c38fd776bc2585a0c3c996e30817f0829fc5064463bdbde0fa2d3d7104.

Toolchain: Lean 4.28.0; Mathlib commit 8f9d9cff6bd728b17a24e163c9402775d9e6a365.

Recorded build: PASS\_CLEAN\_ORIGIN\_RESUMED; public root 8,455 jobs; query roots 8,444 jobs. Eight axiom files contain 42 queries over 35 distinct surfaces, with foundational-only footprints and no sorryAx.

## 2 Gate results

| Gate | Verdict       | Principal evidence                                                     |
|------|---------------|------------------------------------------------------------------------|
| G0   | PASS          | Six hashes; source/package manifests; archive CRC/hash.                |
| G1   | PASS          | Exact split scope, sharpness roles and open chordal scope.             |
| G2   | PASS          | Fresh bounded suites plus E2 ledger and 315,183 exact residual checks. |
| G3   | PASS          | Final/API root closure and all canonical/load-bearing surfaces.        |
| G4   | PASS recorded | Clean-origin resumed logs verified; limitation disclosed.              |
| G5   | PASS          | 51 checks; 144 headings per language; no duplicated long text.         |
| G6   | PASS          | EN 46 / ES 47 pages; clean final logs; all 93 pages inspected.         |
| G7   | PASS internal | Bounded novelty wording; independent E6 confirmation open.             |
| G8   | PASS          | One unambiguous v1.4 target; no residue or duplicate v1.3 artifacts.   |

The executable suite reports **144/144 checks passed** and every gate G0–G8 as PASS.

## 3 External-finding regression

Version 1.4 closes the public-root imports, stale root description, Spanish Proposition 7.4, freeze metadata and reproduction-sequence findings. The recorded author build began clean but resumed after an application restart, so uninterrupted clean-room reproduction remains mandatory externally.

The external auditor receives explicit E2 and E6 hints. E2 must independently rederive the analytic obligations and quantitative bridges; E6 must refresh and confirm the prior-art search and the precise split-versus-chordal novelty scope.

## 4 Overall verdict

**PASS.** No internal blocker, major or minor finding remains. This is not external reproduction, peer review, public-release approval or an independent novelty certificate. The package is ready for residual external adversarial audit.